

Creation Date 27-Sep-2010

Revision Date 18-Oct-2023

Revision Number 10

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                                                   |
|---------------------------|---------------------------------------------------|
| Product Description:      | <u>Zinc chloride</u>                              |
| Cat No. :                 | Z/0850/53; Z/0850/60; Z/0850/68                   |
| Synonyms                  | Zinc (II) Chloride; Zinc dichloride; Zinc butter; |
| Index No                  | 030-003-00-2                                      |
| CAS No                    | 7646-85-7                                         |
| EC No                     | 231-592-0                                         |
| Molecular Formula         | Cl <sub>2</sub> Zn                                |
| REACH registration number | 01-2119472431-44                                  |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Recommended Use                | Laboratory chemicals.                                                                       |
| Sector of use                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| Product category               | PC21 - Laboratory chemicals                                                                 |
| Process categories             | PROC15 - Use as a laboratory reagent                                                        |
| Environmental release category | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| Uses advised against           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

E-mail address begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001-703-527-3887

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

# SAFETY DATA SHEET

Zinc chloride

Revision Date 18-Oct-2023

## Physical hazards

Based on available data, the classification criteria are not met

## Health hazards

Acute oral toxicity  
Skin Corrosion/Irritation  
Serious Eye Damage/Eye Irritation  
Specific target organ toxicity - (single exposure)

Category 4 (H302)  
Category 1 B (H314)  
Category 1 (H318)  
Category 3 (H335)

## Environmental hazards

Acute aquatic toxicity  
Chronic aquatic toxicity

Category 1 (H400)  
Category 1 (H410)

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

## Hazard Statements

H302 - Harmful if swallowed  
H314 - Causes severe skin burns and eye damage  
H335 - May cause respiratory irritation  
H410 - Very toxic to aquatic life with long lasting effects

## Precautionary Statements

P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor/physician

## 2.3. Other hazards

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

# SAFETY DATA SHEET

Zinc chloride

Revision Date 18-Oct-2023

| Component     | CAS No    | EC No             | Weight % | CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567                                                    |
|---------------|-----------|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Zinc chloride | 7646-85-7 | EEC No. 231-592-0 | >95      | Acute Tox. 4 (H302)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>STOT SE 3 (H335)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) |

| Component     | Specific concentration limits (SCL's) | M-Factor | Component notes |
|---------------|---------------------------------------|----------|-----------------|
| Zinc chloride | STOT SE 3 (H335) :: C>=5%             | 10       | -               |

|                           |                  |
|---------------------------|------------------|
| REACH registration number | 01-2119472431-44 |
|---------------------------|------------------|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                            |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                            |
| Skin Contact                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.                                                                                                                                                                                                          |
| Ingestion                          | Do NOT induce vomiting. Immediate medical attention is required. Drink plenty of water. Never give anything by mouth to an unconscious person.                                                                                                                                                                                               |
| Inhalation                         | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician or poison control center immediately. If not breathing, give artificial respiration. |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                             |

### 4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Notes to Physician | Treat symptomatically. |
|--------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

# SAFETY DATA SHEET

Zinc chloride

Revision Date 18-Oct-2023

## Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

## Extinguishing media which must not be used for safety reasons

No information available.

## 5.2. Special hazards arising from the substance or mixture

The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses.

## Hazardous Combustion Products

Zinc, Hydrogen chloride gas.

## 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

Technical Rules for Hazardous Substances (TRGS) 510  
Storage Class (LGK) (Germany)

Class 8B

# SAFETY DATA SHEET

Zinc chloride

Revision Date 18-Oct-2023

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **IRE** - 2021 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority

| Component     | The United Kingdom                                                | European Union | Ireland                                                                 |
|---------------|-------------------------------------------------------------------|----------------|-------------------------------------------------------------------------|
| Zinc chloride | STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr |                | TWA: 1 mg/m <sup>3</sup> 8 hr. fume<br>STEL: 2 mg/m <sup>3</sup> 15 min |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

#### Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values; Workers

| Component                          | Acute effects local (Oral) | Acute effects systemic (Oral) | Chronic effects local (Oral) | Chronic effects systemic (Oral) |
|------------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------------|
| Zinc chloride<br>7646-85-7 ( >95 ) |                            |                               |                              | 0.83 mg/kg bw/day               |

| Component                          | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Zinc chloride<br>7646-85-7 ( >95 ) |                              |                                 |                                | DNEL = 8.3mg/kg bw/day            |

| Component                          | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Zinc chloride<br>7646-85-7 ( >95 ) |                                  |                                     |                                    | DNEL = 1mg/m <sup>3</sup>             |

#### Predicted No Effect Concentration (PNEC)

See values below. Zinc.

| Component                          | Fresh water     | Fresh water sediment          | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)       |
|------------------------------------|-----------------|-------------------------------|--------------------|------------------------------------|--------------------------|
| Zinc chloride<br>7646-85-7 ( >95 ) | PNEC = 20.6µg/L | PNEC = 117.8mg/kg sediment dw |                    | PNEC = 100µg/L                     | PNEC = 35.6mg/kg soil dw |

| Component                          | Marine water   | Marine water sediment        | Marine water intermittent | Food chain | Air |
|------------------------------------|----------------|------------------------------|---------------------------|------------|-----|
| Zinc chloride<br>7646-85-7 ( >95 ) | PNEC = 6.1µg/L | PNEC = 56.5mg/kg sediment dw |                           |            |     |

### 8.2. Exposure controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

# SAFETY DATA SHEET

Zinc chloride

Revision Date 18-Oct-2023

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

## Personal protective equipment

### Eye Protection

Goggles (European standard - EN 166)

### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments                                                                 |
|----------------|-------------------|-----------------|-------------|--------------------------------------------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.5 mm          | EN 374      | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Nitrile rubber | > 480 minutes     | 0.35 mm         |             |                                                                                |
| Natural rubber | > 480 minutes     | 0.5 mm          |             |                                                                                |
| Neoprene       | > 480 minutes     | 0.5 mm          |             |                                                                                |
| PVC            | > 480 minutes     | 0.5 mm          |             |                                                                                |
| Viton (R)      | > 480 minutes     | 0.4 mm          |             |                                                                                |

### Skin and body protection

Long sleeved clothing.

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced: full face mask (DIN EN 136)

**Recommended Filter type:** Particulates filter conforming to EN 143; Type P; White

### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Particle filtering: EN149:2001; or; Half mask: EN140 plus filter, EN 143; White

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

#### Physical State

Solid

#### Appearance

White

#### Odor

Odorless

#### Odor Threshold

No data available

#### Melting Point/Range

293 °C / 559.4 °F

#### Softening Point

No data available

#### Boiling Point/Range

732 °C / 1349.6 °F

#### Flammability (liquid)

Not applicable

Solid

#### Flammability (solid,gas)

No information available

#### Explosion Limits

No data available

# SAFETY DATA SHEET

Zinc chloride

Revision Date 18-Oct-2023

|                                         |                                         |                                   |
|-----------------------------------------|-----------------------------------------|-----------------------------------|
| Flash Point                             | No information available                | Method - No information available |
| Autoignition Temperature                | No data available                       |                                   |
| Decomposition Temperature               | No data available                       |                                   |
| pH                                      | 5                                       | 100 g/L aq.sol                    |
| Viscosity                               | Not applicable                          | Solid                             |
| Water Solubility                        | 432 g/100ml (25°C)                      |                                   |
| Solubility in other solvents            | No information available                |                                   |
| Partition Coefficient (n-octanol/water) |                                         |                                   |
| Vapor Pressure                          | 1.3 mbar @ 428 °C                       |                                   |
| Density / Specific Gravity              | No data available                       |                                   |
| Bulk Density                            | 1,400 - 1,800 kg/m <sup>3</sup> (20 °C) |                                   |
| Vapor Density                           | Not applicable                          | Solid                             |
| Particle characteristics                | No data available                       |                                   |

## 9.2. Other information

|                   |                        |
|-------------------|------------------------|
| Molecular Formula | Cl <sub>2</sub> Zn     |
| Molecular Weight  | 136.29                 |
| Evaporation Rate  | Not applicable - Solid |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

### 10.5. Incompatible materials

Strong bases. Strong acids. Cyanides. Sulfides.

### 10.6. Hazardous decomposition products

Zinc. Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Category 4                                                       |
| Dermal     | Based on available data, the classification criteria are not met |
| Inhalation | Based on available data, the classification criteria are not met |

| Component     | LD50 Oral       | LD50 Dermal | LC50 Inhalation                               |
|---------------|-----------------|-------------|-----------------------------------------------|
| Zinc chloride | 350 mg/kg (Rat) | -           | LC50 <= 1975 mg/m <sup>3</sup> ( Rat ) 10 min |

# SAFETY DATA SHEET

Zinc chloride

Revision Date 18-Oct-2023

- (b) skin corrosion/irritation; Category 1 B
- (c) serious eye damage/irritation; Category 1
- (d) respiratory or skin sensitization;  
Respiratory Based on available data, the classification criteria are not met  
Skin Based on available data, the classification criteria are not met
- (e) germ cell mutagenicity; Based on available data, the classification criteria are not met
- (f) carcinogenicity; Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product
- (g) reproductive toxicity; Based on available data, the classification criteria are not met
- (h) STOT-single exposure; Category 3  
Results / Target organs Respiratory system.
- (i) STOT-repeated exposure; Based on available data, the classification criteria are not met  
Target Organs None known.
- (j) aspiration hazard; Not applicable  
Solid
- Symptoms / effects, both acute and delayed Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component     | Freshwater Fish                             | Water Flea         | Freshwater Algae           |
|---------------|---------------------------------------------|--------------------|----------------------------|
| Zinc chloride | LC50: 0.4-2.2 mg/L/96h<br>(Cyprinus carpio) | EC50: 0.2 mg/L/48h | EC50: 0.027-0.105 mg/L/72h |

| Component     | Microtox | M-Factor |
|---------------|----------|----------|
| Zinc chloride |          | 10       |



# SAFETY DATA SHEET

Zinc chloride

Revision Date 18-Oct-2023

## 12.2. Persistence and degradability

|                                              |                                                                                                                 |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Persistence</b>                           | Soluble in water, Persistence is unlikely, based on information available.                                      |
| <b>Degradability</b>                         | Not relevant for inorganic substances.                                                                          |
| <b>Degradation in sewage treatment plant</b> | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. |

## 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component     | log Pow | Bioconcentration factor (BCF) |
|---------------|---------|-------------------------------|
| Zinc chloride |         | 16000 dimensionless           |

## 12.4. Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

## 12.5. Results of PBT and vPvB assessment

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects **Persistent Organic Pollutant** **Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### **Waste from Residues/Unused Products**

Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

#### **European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### **Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment.

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                                         |                          |
|-----------------------------------------|--------------------------|
| <u>14.1. UN number</u>                  | UN2331                   |
| <u>14.2. UN proper shipping name</u>    | ZINC CHLORIDE, ANHYDROUS |
| <u>14.3. Transport hazard class(es)</u> | 8                        |
| <u>14.4. Packing group</u>              | III                      |

### ADR

FSUZ0850

# SAFETY DATA SHEET

Zinc chloride

Revision Date 18-Oct-2023

**14.1. UN number** UN2331  
**14.2. UN proper shipping name** ZINC CHLORIDE, ANHYDROUS  
**14.3. Transport hazard class(es)** 8  
**14.4. Packing group** III

## IATA

**14.1. UN number** UN2331  
**14.2. UN proper shipping name** ZINC CHLORIDE, ANHYDROUS  
**14.3. Transport hazard class(es)** 8  
**14.4. Packing group** III

**14.5. Environmental hazards** Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component     | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|---------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Zinc chloride | 7646-85-7 | 231-592-0 | -      | -   | X     | X    | KE-35535 | X    | X    |

| Component     | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|---------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Zinc chloride | 7646-85-7 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component     | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Zinc chloride | 7646-85-7 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Seveso III Directive (2012/18/EC)

| Component | CAS No | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|--------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
|-----------|--------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|

# SAFETY DATA SHEET

Zinc chloride

Revision Date 18-Oct-2023

|               |           |                |                |
|---------------|-----------|----------------|----------------|
| Zinc chloride | 7646-85-7 | Not applicable | Not applicable |
|---------------|-----------|----------------|----------------|

**Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals**

Not applicable

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component     | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|---------------|---------------------------------------|-------------------------|
| Zinc chloride | WGK3                                  |                         |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed  
H314 - Causes severe skin burns and eye damage  
H400 - Very toxic to aquatic life  
H410 - Very toxic to aquatic life with long lasting effects  
H318 - Causes serious eye damage  
H335 - May cause respiratory irritation

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

# SAFETY DATA SHEET

Zinc chloride

Revision Date 18-Oct-2023

**PBT** - Persistent, Bioaccumulative, Toxic

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Training Advice**

Chemical incident response training.

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Creation Date** 27-Sep-2010

**Revision Date** 18-Oct-2023

**Revision Summary** Not applicable.

**This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended.**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**